

A. Installing Multiple R Versions from Source

1. Install Prerequisites

```
bash
sudo apt update
sudo apt install -y build-essential libcurl4-openssl-dev libssl-dev libxml2-dev gdebi-core
```

2. Download and Compile Each R Version

Create directories and download R source code:

```
bash
cd /opt
sudo mkdir -p R-4.2.3 R-4.3.1
sudo chown $(whoami):$(whoami) R-4.2.3 R-4.3.1
```

```
cd /tmp
wget https://cran.r-project.org/src/base/R-4/R-4.2.3.tar.gz
wget https://cran.r-project.org/src/base/R-4/R-4.3.1.tar.gz
Build and install each version:
```

```
bash
# Install R 4.2.3
tar -xzf R-4.2.3.tar.gz
cd R-4.2.3
./configure --prefix=/opt/R-4.2.3 --enable-R-shlib
make -j$(nproc)
sudo make install
```

```
# Install R 4.3.1
cd /tmp
tar -xzf R-4.3.1.tar.gz
cd R-4.3.1
./configure --prefix=/opt/R-4.3.1 --enable-R-shlib
make -j$(nproc)
sudo make install
```

3. Switch Between Versions (Command Line)

```
bash
export PATH=/opt/R-4.2.3/bin:$PATH
R --version
```

B. Install and Configure RStudio Server

1. Install RStudio Server

Download from [Posit's website](https://posit.co/what-is-rstudio-server/):

```
bash
wget https://download2.rstudio.org/server/jammy/amd64/rstudio-server-2023.06.2-561-amd64.deb
sudo gdebi rstudio-server-2023.06.2-561-amd64.deb
```

2. Configure RStudio Server for Multiple R Versions

Create or edit /etc/rstudio/r-versions:

```
bash
sudo nano /etc/rstudio/r-versions
```

Add both R installations:

```
Code
/opt/R-4.2.3
```

/opt/R-4.3.1

Save and restart RStudio Server:

bash

sudo rstudio-server restart

C. Usage

- **Command line:** Use `export PATH=/opt/R-X.X.X/bin:$PATH` to switch versions
- **RStudio Server:** Users will see an R version dropdown in the UI to select their preferred version
- **Access:** Navigate to `http://your-server:8787` in your browser